

XIAOYUN XU

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EDUCATION

Radboud University, Nijmegen, The Netherlands

May, 2022 - Jan, 2026

PhD in Computer Science

Supervisor: Dr. Stjepan Picek

Thesis: No Time to Spare: Adversarial Machine Learning at Training and Inference Time

University of Bristol, Bristol, UK

2017 - 2018

MSc in Advanced Computing

Thesis: Investigating the effectiveness of existing machine-learning-based compiler optimization techniques

University of Electronic Science and Technology of China

2013 - 2017

BEng in Software Engineering

Thesis: Vehicle license plate recognition based on SVM and ANN

RESEARCH PUBLICATION

1. **Xiaoyun Xu**, Shujian Yu, Zhuoran Liu, and Stjepan Picek. MIMIR: Masked Image Modeling for Mutual Information-based Adversarial Robustness. *Accepted by The Network and Distributed System Security (NDSS)*, 2026.
2. **Xiaoyun Xu**, Zhuoran Liu, Stefanos Koffas, and Stjepan Picek. Towards Backdoor Stealthiness in Model Parameter Space. *ACM Conference on Computer and Communications Security (CCS)*, 2025.
3. **Xiaoyun Xu**, Zhuoran Liu, Stefanos Koffas, Shujian Yu, and Stjepan Picek. BAN: Detecting Backdoors Activated by Adversarial Neuron Noise. *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
4. Zhuoran Liu, Senna van Hoek, Péter Horváth, Dirk Lauret, **Xiaoyun Xu**, and Lejla Batina. Real-world Edge Neural Network Implementations Leak Private Interactions Through Physical Side Channel. *arXiv preprint*, 2025.
5. **Xiaoyun Xu**, Oguzhan Ersoy, Behrad Tajalli, and Stjepan Picek. Universal Soldier: Using universal adversarial perturbations for detecting backdoor attacks. *IEEE/IFIP International Conference on Dependable Systems and Networks Workshops (DSN-W)*, 2024.
6. **Xiaoyun Xu**, and Stjepan Picek. Poster: Boosting Adversarial Robustness by Adversarial Pre-training. *ACM Conference on Computer and Communications Security (CCS)*, 2023.
7. **Xiaoyun Xu**, Guilherme Perin, and Stjepan Picek. IB-RAR: Information Bottleneck as Regularizer for Adversarial Robustness. *IEEE/IFIP International Conference on Dependable Systems and Networks Workshops (DSN-W)*, 2023.
8. **Xiaoyun Xu**, Jingzheng Wu, Mutian Yang, Tianyue Luo, Qianru Meng, Weiheng Li, and Yanjun Wu. AI-CTO: Knowledge graph for automated and dependable software stack solution. *Journal of Intelligent and Fuzzy Systems*, 2021.
9. **Xiaoyun Xu**, Jingzheng Wu, Mutian Yang, Tianyue Luo, Xu Duan, Weiheng Li, Yanjun Wu, and Bin Wu. Information leakage by model weights on federated learning. *In Proceedings of the 2020 workshop on privacy-preserving machine learning in practice, CCS workshop PPLMP*, 2020.
10. **Xiaoyun Xu**, Jingzheng Wu, Mutian Yang, Tianyue Luo, A method of shortening vulnerability attack window based on knowledge graph reasoning, CN110378126B, (Patent)
11. Vulnerability management platform based on Knowledge Graph, 2019SR0860641, (Software Copyright)

RESEARCH EXPERIENCE AND PROJECTS

Horizon Europe – the Framework Programme for Research and Innovation	Nov, 2025 - Present
• Project Title: Secure Hardware and Software for AI systems (SHASAI) • Grant agreement ID: 101225866	
Dutch national e-infrastructure (SURF Cooperative)	Jan, 2025 - Dec, 2025
• Project Title: LLM pruning against backdoor attacks • grant no. EINF-12068	
Dutch national e-infrastructure (SURF Cooperative)	Sep, 2024 - Sep, 2025
• Project Title: Adversarially train the projection layer for robustness of vision and language alignment • grant no. EINF-10853	
Institute of Software, Chinese Academy of Sciences	2018 - 2020
• Research Assistant, topics: Knowledge Graph, security vulnerabilities, and Interpretable AI • Domain knowledge graph of software vulnerabilities	

ACADEMIC SERVICES

Reviewers/PC	CCS, BMVC, ICLR, NeurIPS, SaTML
External Reviewers	S&P, NDSS, USENIX Security

TEACHING

- Guest Lecturer, Academic Writing and Research Methods (NWI-IBC043), 2025, Radboud University
- Teaching Assistant, Security and Privacy of Machine Learning (NWI-IMC069), 2023, Radboud University

CAMPUS EXPERIENCE

Summer School, Šibenik, Croatia	2022.06
• Topic: On real-world crypto and privacy	
Exchange program at Christopher Newport University, America	2015.07 – 2015.08
• Received 10,000 (Chinese yuan) scholarship • Explored the American education model and culture	
Exchange program at Waseda University, Japan	2016.01 – 2016.02
• Received a scholarship of 50,000 yen (JPY) • Learned more about Japanese culture and visited Japanese well-known enterprises	

SKILLS

Languages	Chinese, English.
Programming	C/C++, Python, Linux, HTML, Git, CSS, Makefile, NodeJS, JavaScript, Neo4j, Cypher.
Courses	Computational Neuroscience, Cloud Computing, Computer Graphics, Data Structures and Algorithms, Database Principles and Applications, Computer Networks, Image Processing, Computer Vision, etc.